PayPilot AI — Project Documentation

An Autonomous Fintech Payment Platform Powered by AI & Official PayPal APIs

1. What is PayPilot AI? (In Simple Words)

PayPilot AI is an intelligent financial co-pilot built for modern businesses, agencies, and startups. Managing customer invoices, paying freelance contractors, and updating transaction ledgers normally requires tedious manual work across multiple banking dashboards.

PayPilot AI automates these operations. You can simply tell the AI: *"Create an invoice for $250 for website design"* or *"Pay $120 to Elena for backend tasks"*. The system understands your intent, applies strict financial safety rules, creates the verified transaction, and executes the money transfer safely using official **PayPal Sandbox APIs**.

Key Takeaway: PayPilot AI acts like a smart financial assistant that turns everyday English instructions into safe, audit-backed PayPal orders, contractor payouts, and real-time ledger accounting.

2. Core Features & What Each Section Does

Feature / Module	What It Does (Simple Explanation)	Real-World Benefit
AI Copilot & Assistant /agent	Understands natural conversational English. Extracts amounts, recipient names, currencies, and proposes a structured payment plan.	No need to click through complicated banking forms. Just speak or type.
Invoice Management /invoices	Creates custom bills, tracks payment statuses (Sent, Paid, Overdue), and provides instant PayPal checkout buttons.	Clients can pay immediately via PayPal with 1-click reconciliation.
Contractor Payouts /payouts	Stages disbursements to freelancers. Includes automated duplicate payment checks and approval gates for transactions over $100.	Prevents accidental double payments and gives managers full approval control.
Workflow Automation /workflows	A visual 'If-This-Then-That' engine. For example: *When invoice is paid -> automatically update ledger and send receipt*.	Eliminates repetitive manual bookkeeping and follow-up tasks.
Ledger & Reporting /transactions	A double-entry financial journal recording every incoming payment and outgoing payout. Includes 1-click CSV download.	Clean, tax-ready and audit-ready records for accountants.
Contractors Directory Saved Payees	Directory of verified clients and vendors with their validated PayPal emails and categories.	Speedy payout staging without remembering or retyping emails.

3. Built-in Safety & Security Guardrails

To protect business funds, PayPilot AI follows strict financial rules:

• **Human-in-the-Loop Approval:** Any payout exceeding $100 requires an operator to click 'Approve' before funds can leave the account.
• **Double-Payment Prevention (Idempotency):** If someone accidentally requests the same payout twice within 24 hours, the system flags it as a duplicate.
• **Append-Only Audit Trail:** Every proposal, approval, rejection, and payment is permanently recorded in the system audit log.
• **Input Sanitization:** All data entries are sanitized against cross-site scripting (XSS) and injection threats.

4. How It Works Under the Hood

1. Frontend (UI)	Built with **React & TypeScript**. Fast, modern, responsive interface with live loading skeletons, instant toast alerts, and modal dialogs.
2. Backend (API)	Powered by **Python & FastAPI**. Validates all inputs using strict Pydantic schemas, runs the AI intent parser, and applies safety business logic.
3. PayPal & Database	Connects directly to **PayPal Sandbox Orders & Payouts API**. Persists all records in a high-reliability local **SQLite database**.

5. Quick Start Guide: How to Use It

1. **Access the Dashboard:** Open your browser to http://localhost:5173.
2. **Seed Demo Data:** Click **'Seed Demo Data'** at the top right to populate sample invoices, payouts, and charts.
3. **Try the AI Assistant:** Go to **AI Agent** and click one of the suggested prompts or type: *'Create an invoice for $250'*.
4. **Approve & Disburse:** Head to **Contractor Payouts** to review staged payouts and click **Disburse Funds**.
5. **Inspect Ledger & Export:** Visit **Transactions** and click **Export CSV** to download your accounting spreadsheet.